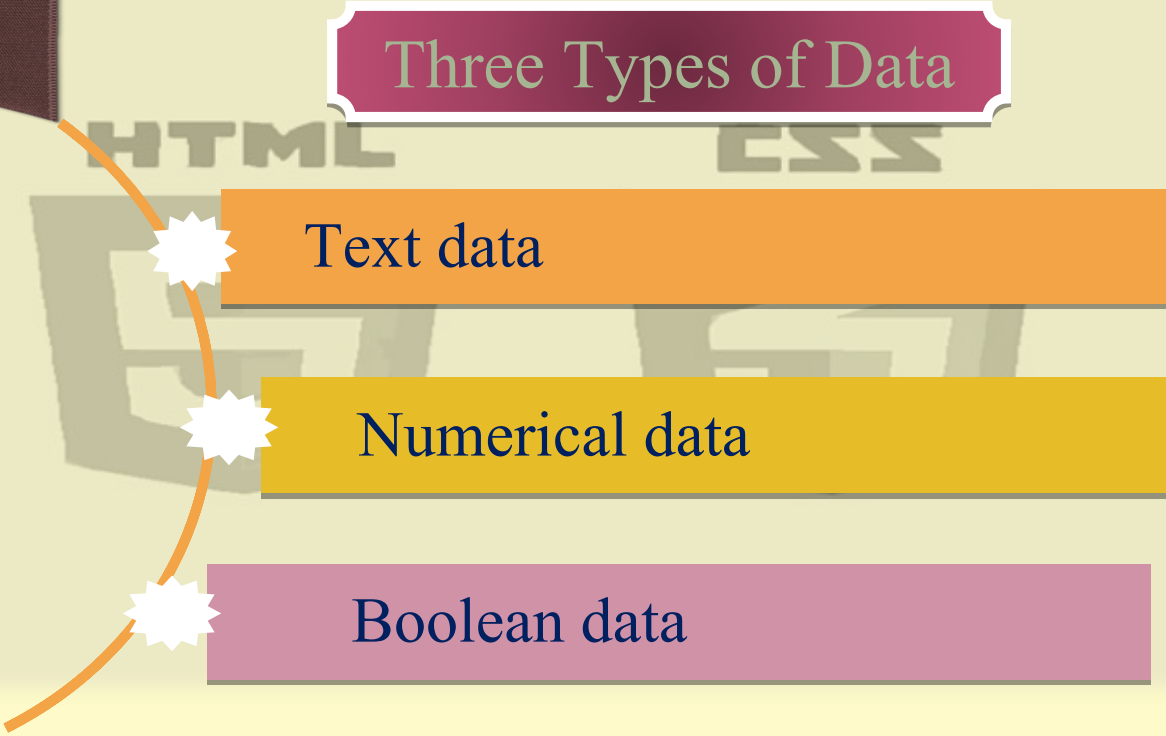


7-2 Usage of Data Types and Variables

Three Types of Data



Text data

Numerical data

Boolean data

Types of Data

Text Data

One or more characters of text is a string

Literal strings using ' or " characters

For example:

- ✓ "Hello World" or ' Hello World'
- ✓ "Peter O' Toole" → Peter O' Toole
- ✗ 'Peter O' Toole' -error
- 👉 (\)-backslash character-escape character
- ✓ 'Peter O \'Toole' → Peter O' Toole
- ✓ ' Hello "Paul" ' → Hello "Paul"
- ✗ "Hello "Paul"" -error

Types of Data

Numerical Data

- Whole numbers-such as 145, Integers (positive or negative), -2^{53} to 2^{53}
- Fractional numbers-such as 1.234, also known as *floating-point numbers*

Boolean Data

- Two values:
 - *true* for yes and
 - *False for no*

Escape Sequences

- ❖ there are instances where you can't use a character directly but must use an escape sequence.

Escape Sequences	Character Represented
\b	Backspace
\f	Form feed
\n	New Line
\r	Carriage return
\t	Tab
\'	Single quote
\"	Double quote
\\	Backslash
\xNN	NN is a hexadecimal number that identifies a character in the Latin-1 character set

"\xA9 Paul Wilton".

© Paul Wilton.

OK

Variables

★ Variable names are **case sensitive**.

For example:

✗ **myVariable** is **not the same** as **myvariable**

✗ Can't use **Reserved words** (e.g **var** or **with**)

✗ Can't use **&** and **%**

➡ Allow to use numbers ,

✗ but the names must not begin with numbers

Variables Examples

 Examples of valid and invalid variable names:

Invalid names	Valid names
99variables	myVariable99
My%Variable	myPercent Variable
theGood&theBad	the_Good_and_the_Bad



Constant, variable

To define constant	const constant_name = constant value;
To define variable	var variable_name; var name1 = initial_value1, name2 = initial_value2, ;

Scope of the variable

- ★ Variables can be **global** (when you declare outside of any function) or can be **local** within a function.

Creating Variables and Giving Them Values

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN→  
→http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
```

```
<html xmlns="http://www.w3.org/1999/xhtml">
```

```
<head> </head>
```

```
<body>
```

```
<script type="text/javascript">
```

```
var myFirstVariable;
```

```
myFirstVariable = →Hello";
```

```
alert(myFirstVariable); //Hello
```

```
myFirstVariable = 54321;
```

```
alert(myFirstVariable); //54321
```

```
</script>
```

```
</body>
```

```
</html>
```



Assigning Variables with the Value of Other Variables

- JS can assign a variable with a number or string
- JS can also assign a variable with the data stored inside another variable

```
var myVariable;  
var myOtherVariable;  
myOtherVariable = 22;
```

- 👉 the following line assign **myVariable** with the value of other variable **myOtherVariable**(that is, 22)

```
myVariable = myOtherVariable;
```

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"  
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
```

```
<html xmlns="http://www.w3.org/1999/xhtml">
```

```
<body>
```

```
<script language="JavaScript" type="text/javascript">
```

```
var string1 = "Hello";
```

```
var string2 = "Goodbye";
```

variable declaration

```
alert(string1); // Hello
```

```
alert(string2); // Goodbye
```

```
string2 = string1;
```

Assigning variable with the value of other variable

```
alert(string1); // Hello
```

```
alert(string2); // Hello
```

Assigning variable with new value

```
string1 = "Now for something different";
```

```
alert(string1); // Now for something different
```

```
alert(string2); // Hello
```

```
</script>
```

```
</body> </html>
```